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United States Patent	12415585
Kind Code	B2
Date of Patent	September 16, 2025
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Vehicle information notification device

Abstract

This vehicle information notification device includes: movable portions which are provided in at least one of a pair of left and right vehicle body outer portions in a vehicle body, steerably supporting a steering system component, to be operated by actuators, wherein the movable portions include outer surfaces which are able to contact a driver's body, and wherein the movable portions notify the driver of predetermined information by operating the outer surfaces in a predetermined motion.

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Appl. No.:	18/034712
Filed (or PCT Filed):	October 18, 2021
PCT No.:	PCT/JP2021/038429
PCT Pub. No.:	WO2022/107527
PCT Pub. Date:	May 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230399071 A1	Dec. 14, 2023

Foreign Application Priority Data

JP 2020-192716 Nov. 19, 2020

Publication Classification

Int. Cl.: B62J50/23 (20200101); B62J50/22 (20200101); B62K21/00 (20060101)

U.S. Cl.:

CPC B62J50/23 (20200201); B62J50/22 (20200201); B62K21/00 (20130101);

Field of Classification Search

CPC: B62J (6/24); B62J (25/02); B62J (27/00); B62J (45/41); B62J (50/22); B62J (50/23); B60W (10/18); B60W (10/20); B60W (30/18163); B60W (50/14); B60W (50/16); B60W (2300/36); B60W (2050/146); B60Q (9/008); B60T (8/1706); B62D (1/14); B62K (21/26)

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Background/Summary

TECHNICAL FIELD

- (1) The present invention relates to a vehicle information notification device.
- (2) The present application claims priority based on Japanese Patent Application No. 2020-192716 filed on Nov. 19, 2020, the contents of which are incorporated herein by reference.

BACKGROUND ART

- (3) Conventionally, it is known that a tactile stimulus device provided on a rim of a steering wheel or the like gives information to a driver through the tactile sense during automatic driving of a vehicle (for example, see Patent Document 1).

RELATED ART DOCUMENT

Patent Document

- (4) Patent Document 1: Japanese Unexamined Patent Application, First Publication No. 2018-181269

SUMMARY

Problems to be Solved by the Invention

- (5) Since the above-described tactile information notification is performed on the part that the driver constantly touches while driving the vehicle compared to the visual information notification, it is possible to reliably notify the driver of information.
- (6) However, when the driver needs to perform the steering operation while receiving notified information, it is conceivable that the driver will feel complicated when receiving the notified information. Accordingly, there is a desire to improve the complicated feeling.
- (7) Here, an object of the present invention is to easily receive information notified in a vehicle information notification device that notifies a driver of information by tactile sense.

Means for Solving the Problem

- (8) As a solution to the above problems, a vehicle information notification device of the present invention includes: movable portions which are provided in at least one of a pair of left and right vehicle body outer portions in a vehicle body, steerably supporting a steering system component, to be operated by actuators. The movable portions include outer surfaces which are able to contact a driver's body, and the movable portions notify the driver of predetermined information by operating the outer surfaces in a predetermined motion.
- (9) According to this configuration, the movable portion is provided in at least one of the pair of left and right vehicle body outer portions of the vehicle body. By the operation of the movable

portion, predetermined information can be notified to the driver's body in advance. The movable portion needs to be separated from the steering system component (especially, handle).

Accordingly, the notification of information by the movable portion and the steering operation are separated from each other and the driver can easily receive notified information.

(10) In the present invention, the movable portions may include the outer surfaces provided in front of a seat and above a step to be able to contact driver's legs.

(11) According to this configuration, an outer surface that can contact the driver's legs is placed in front of the seat and above the step (where the driver performs a knee gripping operation) to be operated. Accordingly, predetermined information can be notified to the driver's legs.

(12) In the present invention, the movable portions may be operated so that the outer surfaces are moved outward in a vehicle width direction.

(13) According to this configuration, the movable portion is operated so that the outer surface is moved outward in the vehicle width direction. Accordingly, it is possible to increase the certainty of contact with the driver's legs and to easily receive notified information by the driver.

(14) In the present invention, the movable portions may be respectively provided in the pair of left and right vehicle body outer portions and are operable independently of each other.

(15) According to this configuration, since the pair of left and right movable portions is operated independently of each other, it is possible to combine the difference in which of the left and right movable portions is operated. Accordingly, it is possible to notify the driver of a lot of information and to increase the type of information to be notified.

(16) In the present invention, the vehicle information notification device may include an operation intervention function capable of intervening in operations performed by the driver and the movable portions may be operated while being interlocked with the operation intervention function.

(17) According to this configuration, the movable portion is operated while being interlocked with the operation intervention function for the operation of the driver. Accordingly, it is possible to notify the driver that the operation intervention function is operated by the operation of the movable portion.

(18) In the present invention, when the operation intervention function is operated, the movable portions may start to be operated at a timing earlier than a timing of starting the operation of the operation intervention function.

(19) According to this configuration, the movable portion starts to be operated at a timing earlier than the start of the operation of the operation intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the operation intervention function in advance.

(20) In the present invention, at least one of a steering intervention function and a braking intervention function may be provided as the operation intervention function.

(21) According to this configuration, at least one of the steering intervention function and the braking intervention function of the operation intervention function is operated. Accordingly, it is possible to notify the driver by the operation of the movable portion.

(22) In the present invention, both the left and right movable portions may be operated when the braking intervention function is operated.

(23) According to this configuration, the operation of the braking intervention function is notified to the driver by the operation of both left and right movable portions. Accordingly, it is possible to increase the certainty of the notification of the operation of the braking intervention function that tends to cause behavior.

(24) In the present invention, one of the left and right movable portions may be operated when the steering intervention function is operated.

(25) According to this configuration, the operation of the steering intervention function is notified to the driver by the operation of one of the left and right movable portions. Accordingly, it is possible to notify the driver of information including a steering direction by the steering intervention function depending on which of the left and right movable portions is operated.

(26) In the present invention, the steering intervention function may include a course change function for changing a course of a vehicle to any one of left and right. The movable portions may be operated so that the outer surfaces are moved outward in the vehicle width direction. When the steering intervention function is operated, the movable portions that change the course of the vehicle may be operated so that the outer surfaces are moved outward in the vehicle width direction.

(27) According to this configuration, the movable portion that changes the course of the vehicle is operated so that the outer surface is moved outward in the vehicle width direction when the course change function of the steering intervention function is operated. Accordingly, the leg of the driver on the side of the course change is opened outward in the vehicle width direction, and the posture suitable for steering can be urged.

(28) In the present invention, in a standby state in which a system of the operation intervention function is activated and the operation intervention function is not operated yet, the movable portions may be operated as below. That is, the movable portions may be operated in an operation mode different from that when the operation intervention function is operated.

(29) According to this configuration, there is a standby state of the operation intervention function before the operation intervention function is operated and in the standby state, the movable portion is operated in an operation mode different from that when the operation intervention function is operated. Accordingly, it is possible to notify the driver that the operation intervention function is in the standby state.

(30) In the present invention, the vehicle information notification device may further include: display devices which display predetermined information for the driver while being interlocked with the operation of the movable portions.

(31) According to this configuration, the vehicle information notification device further includes the display device which is operated while being interlocked with the operation of the movable portion. Accordingly, it is possible to further reliably notify the driver of information by the operation of the movable portion and the display of the display device.

Advantage of the Invention

(32) According to the present invention, in the vehicle information notification device that notifies the driver of information by tactile sense, information can be easily notified.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a left side view of a motorcycle of an embodiment of the present invention.

(2) FIG. 2 is a perspective view of a front part of a vehicle body of the motorcycle when viewed from a driver's line of sight.

(3) FIG. 3 is an explanatory diagram of a steering actuator of the motorcycle.

(4) FIG. 4 is a block diagram of a driving assistance device of the motorcycle.

(5) FIG. 5 is a cross-sectional view taken along a line V-V of FIG. 1 and shows a non-operation state of a left movable portion.

(6) FIG. 6 is a cross-sectional view corresponding to FIG. 5 and showing a first operation state of the left movable portion.

(7) FIG. 7 is a cross-sectional view corresponding to FIG. 5 and showing a second operation state of the left movable portion.

(8) FIG. 8 is an explanatory diagram of an operation in a standby state of the driving assistance device.

(9) FIG. 9 is a graph showing temporal changes in the operation of various devices in the standby state of the driving assistance device.

(10) FIG. 10 is an explanatory diagram of an operation when a braking intervention operation of the driving assistance device is operated.

(11) FIG. 11 is a graph showing temporal changes in the operation of various devices when the braking intervention operation of the driving assistance device is operated.

(12) FIG. 12 is an explanatory diagram of an operation when a left steering intervention operation of the driving assistance device is operated.

(13) FIG. 13 is a graph showing temporal changes in the operation of various devices when the left steering intervention operation of the driving assistance device is operated.

(14) FIG. 14 is an explanatory diagram of an operation when a right steering intervention operation of the driving assistance device is operated.

(15) FIG. 15 is a graph showing temporal changes in the operation of various devices when the right steering intervention operation of the driving assistance device is operated.

DESCRIPTION OF THE EMBODIMENTS

(16) Hereinafter, an embodiment of the present invention will be described with reference to the drawings. In the following description, directions such as front, rear, left, and right are the same as the directions of the vehicle described below unless otherwise specified. An arrow FR indicating the front of the vehicle, an arrow LH indicating the left of the vehicle, and an arrow UP indicating the upper side of the vehicle are shown at appropriate locations in the drawings used in the following description.

(17) <Whole Vehicle>

(18) FIG. 1 shows a motorcycle 1 as an example of a vehicle of the embodiment. The motorcycle 1 includes a front wheel (steered wheel) 3 which is steered by a handle 2 and a rear wheel (driving wheel) 4 which is driven by a power unit 20. The motorcycle 1 is a saddle type vehicle in which a driver straddles a vehicle body. The motorcycle 1 is able to swing (bank) the vehicle body in the left and right direction (roll direction) based on ground contact points of the front and rear wheels 3 and 4. The vehicle of the embodiment is not limited to a vehicle that turns in a direction of banking a vehicle body like a motorcycle. The vehicle of the embodiment includes a vehicle that turns by steering a steered wheel without banking the vehicle body.

(19) The motorcycle 1 includes a steering system component 10A having a handle 2 and a front wheel 3. The steering system component 10A is supported by a head pipe 6 to be steerable. The head pipe 6 is located at a front end portion of a vehicle body frame 5 that forms the skeleton of the motorcycle 1. The front wheel 3 is supported by lower end portions of a pair of left and right front forks 10 of the steering system component 10A. The vehicle body frame 5 is surrounded by a vehicle body cover 12. In the motorcycle 1, a vehicle body including a vehicle body frame 5 and the like (a vehicle body steerably supporting the steering system component 10A) is referred to as a vehicle body 1A.

(20) The vehicle body frame 5 includes the head pipe 6, a pair of left and right main frames 7, a pair of left and right pivot frames 8, and a pair of left and right seat frames 9.

(21) The head pipe 6 steerably supports the steering system component 10A. The left and right main frames 7 extend rearwardly downward from the head pipe 6. The left and right pivot frames 8 extend downward from the rear end portions of the left and right main frames 7 respectively. The left and right seat frames 9 extend rearwardly upward from the respective upper portions of the left and right pivot frames 8.

(22) A pivot shaft 8a extending in the vehicle width direction is provided between the left and right pivot frames 8. The front end portion of the swing arm 11 is supported by the left and right pivot frames 8 through the pivot shaft 8a to swing up and down. The rear wheel 4 is supported by the rear end portion of the swing arm 11. A cushion unit (not shown) serving as a shock absorber is provided between the vehicle body frame 5 and the swing arm 11.

(23) A fuel tank 18 is supported by the upper portions of the left and right main frames 7. A seat 19 is supported behind the fuel tank 18 by the left and right seat frames 9. A pair of left and right steps

25 on which the feet of the driver seated on the seat 19 are placed are arranged below the seat 19.

(24) The power unit 20 of the motorcycle 1 is supported by the left and right main frames 7 and the left and right pivot frames 8. An output shaft of the power unit 20 is connected to the rear wheel 4 through a chain transmission mechanism (not shown) so that power can be transmitted.

(25) The power unit 20 integrally includes an engine (internal combustion engine) 13 as a prime mover and a transmission 21 connected to the rear of the engine 13.

(26) The motorcycle 1 includes a front wheel brake 3B which brakes the front wheel 3 and a rear wheel brake 4B which brakes the rear wheel 4. Each of the front wheel brake 3B and the rear wheel brake 4B is a disc brake.

(27) The front wheel brake 3B and the rear wheel brake 4B appropriately brake the rotation of the front wheel 3 and the rear wheel 4 by operating a brake lever 43 and a brake pedal which are brake operators. Further, the front wheel brake 3B and the rear wheel brake 4B appropriately brake the rotation of the front wheel 3 and the rear wheel 4 by operating a brake actuator 102 (see FIG. 4) which will be described later.

(28) The motorcycle 1 includes a driving assistance device 70 (see FIG. 4) that assists the driver's driving operation (in the embodiment, the steering operation for steering the front wheel 3 and the braking operation for braking the front wheel 3 and the rear wheel 4). The driving assistance device 70 includes a control device 71 that controls a function of automatically intervening in the driver's driving operation (automatic operation intervention function). The automatic operation intervention function includes an automatic steering intervention function and an automatic braking intervention function. The driving assistance device 70 will be described later.

(29) FIG. 2 shows the front part of the vehicle body when viewed from the driver's line of sight. The upper portions of the left and right front forks 10 are supported by the head pipe 6 through a steering stem 31. The left and right front forks 10 are telescopic shock absorbers. The steering stem 31 includes a top bridge 32 and a bottom bridge 33 which connect the upper portions of the left and right front forks 10 and a stem shaft (steering shaft) 34 which is inserted through the head pipe 6. The front part of the vehicle body is covered with a front cowl 27 of the vehicle body cover 12.

(30) For example, the handle 2 of the motorcycle 1 is separate left and right handles, and includes a pair of left and right handles 36 and 37. For example, the right handle 36 and the left handle 37 are attached to the upper portions of the left and right front forks 10 below the top bridge 32, respectively.

(31) Referring to FIGS. 1 and 5, the vehicle body 1A includes a pair of left and right vehicle body outer portions 1L and 1R which faces the outside in the vehicle width direction. The left and right vehicle body outer portions 1L and 1R are provided with movable portions 56 and 57 which are operated by actuators 105L and 105R. The left and right movable portions 56 and 57 are arranged in front of the seat 19 and above the step 25. The driver's legs, which are seated on the seat 19 and put their feet on the steps 25, extend to the left and right outside of the fuel tank 18. For the legs, knees are bent on the left and right outside of the fuel tank and the lower legs are extended toward the steps 25. The movable portions 56 and 57 are arranged to overlap the legs when viewed from the side of the vehicle. The outer side surfaces (outer surfaces) 56e and 57e of the movable portions 56 and 57 facing outward in the vehicle width direction can come into contact with the driver's legs from the inside in the vehicle width direction.

(32) The movable portions 56 and 57 function as means for notifying the driver's legs of predetermined information by operating the outer side surfaces 56e and 57e in a predetermined motion. In the embodiment, the movable portions 56 and 57 function as means for notifying the driver of information on the automatic operation intervention function. The movable portions 56 and 57 are operated to move the outer side surfaces 56e and 57e outward in the vehicle width direction. Each of the movable portions 56 and 57 is included in a leg pressing device 72 which presses the driver's leg inward in the vehicle width direction.

(33) The leg pressing device 72 includes a right leg pressing portion 72R which is provided in the

right vehicle body outer portion **1R**, a left leg pressing portion **72L** which is provided in the left vehicle body outer portion **1L**, and a leg pressing control unit **89** to be described later (see FIG. **4**). The right leg pressing portion **72R** and the left leg pressing portion **72L** can move (swing) a right movable portion (movable portion) **56** and a left movable portion **57** each having a plate shape along the outer side surface of the vehicle body in the vehicle width direction. The right movable portion **56** and the left movable portion **57** extend obliquely rearwardly downward along the upper edge of the main frame **7** when viewed from the side of the vehicle. The right movable portion **56** and the left movable portion **57** can contact the left and right legs (thighs, knees, and the like) of the driver from the inside in the vehicle width direction.

(34) A position in which the right movable portion **56** and the left movable portion **57** are arranged in the vehicle body **1A** is a position in which the driver sandwiches the vehicle body **1A** with the left and right legs (so-called knee gripping position). The right vehicle body outer portion **1R** and the left vehicle body outer portion **1L** are provided with a pair of left and right leg grip portions **120** formed to align the left and right legs of the driver. The right movable portion **56** and the left movable portion **57** are respectively arranged at the left and right leg grip portions **120**.

(35) Thus, when the driver performs a knee gripping operation in a normal riding posture, the left and right legs of the driver naturally come into contact with the outer side surfaces **56e** and **57e** of the right movable portion **56** and the left movable portion **57**. The right movable portion **56** and the left movable portion **57** are operated from an initial position **P0** (see FIG. **5**) to operation positions **P1** and **P2** (see FIGS. **6** and **7**). The initial position **P0** is a position in which the outer side surfaces **56e** and **57e** are arranged along the outer side surfaces of the vehicle body outer portions **1R** and **1L**. The operation positions **P1** and **P2** are positions in which the outer side surfaces **56e** and **57e** are moved (swung) outward in the vehicle width direction. When the right movable portion **56** and the left movable portion **57** are located at the operation positions **P1** and **P2**, the left and right outer side surfaces **56e** and **57e** protrude outward in the vehicle width direction from the remaining general surfaces of the leg grip portion **120**.

(36) The operation positions **P1** and **P2** include the first operation position **P1** (see FIG. **6**) and the second operation position **P2** (see FIG. **7**). The first operation position **P1** is a position in which the outer side surfaces **56e** and **57e** are moved (swung) outward in the vehicle width direction by a first prescribed amount **S1** from the initial position **P0**. The second operation position **P2** is a position in which the outer side surfaces **56e** and **57e** are further moved (swung) outward in the vehicle width direction by a second prescribed amount **S2** from the first operation position **P1**. The right movable portion **56** and the left movable portion **57** move (swing) outward in the vehicle width direction so that the driver's legs are pressed from the inside in the vehicle width direction. This pressing notifies the driver of information such as that the automatic operation intervention function is operating.

(37) Referring to FIG. **2**, a meter device **61** is disposed in front of a front fork **10**. The meter device **61** is supported by the vehicle body frame **5** or a front cowl **27**. The meter device **61** includes a display screen **62** such as a liquid crystal display which displays images of a vehicle speed and an engine rotation speed and an indicator lamp group **63** which is arranged around the display screen **62** and notifies various information.

(38) The indicator lamp group **63** includes a right indicator lamp (display device) **66** which is disposed on the right side of the display screen **62** and a left indicator lamp (display device) **67** which is disposed on the left side of the display screen **62**. The right indicator lamp **66** emits light while being interlocked with the operation of the right movable portion **56**. The left indicator lamp **67** emits light while being interlocked with the operation of the left movable portion **57**. The display screen **62** notifies the driver of predetermined information by displaying a predetermined image. The indicator lamp group **63** notifies the driver of predetermined information in advance by performing predetermined light-emitting display (lighting or blinking).

(39) FIG. **3** shows the periphery of the top bridge **32** when viewed from above in the axial direction

of the stem shaft **34**.

(40) A steering input is automatically applied to the steering system component **10A** by a steering assist device **73** separately from the operation of the handle **2** by the driver.

(41) The steering assist device **73** includes a steering actuator **74**, an arm **75**, a connecting rod **76**, and a steering control unit **77**.

(42) The steering actuator **74** includes an electric motor which is a drive source of the automatic steering intervention function. The steering actuator **74** is fixed to, for example, the vehicle body frame **5**. A base end portion of the arm **75** is fixed to a drive shaft **74a** which is an output shaft of the steering actuator **74** to be rotatable together. One end portion of the connecting rod **76** is swingably connected to the tip portion of the arm **75** through a first connecting pin **78**. The other end portion of the connecting rod **76** is swingably connected to a rod connecting portion **32a** provided on the top bridge **32** through a second connecting pin **79**.

(43) The operation of the steering actuator **74** is controlled by the steering control unit **77**. The output of the steering actuator **74** (the rotation torque of the drive shaft **74a**) is transmitted to the top bridge **32** through the arm **75** and the connecting rod **76**. Accordingly, the steering actuator **74** generates a steering torque (assist torque) in the steering system component **10A**.

(44) <Driving Assistance Device>

(45) As shown in FIG. **4**, the driving assistance device **70** includes a control device **71**, various sensors **81**, and various devices **82**. The control device **71** controls the operation of various devices **82** based on the detection information acquired from various sensors **81**.

(46) The control device **71** is composed of, for example, a single or a plurality of electronic control units (ECUs). At least a part of the control device **71** may be realized by cooperation of software and hardware.

(47) The control device **71** includes an engine control unit **85**, a steering control unit **77**, a brake control unit **87**, a display control unit **88**, and a leg pressing control unit **89**.

(48) Various sensors **81** include a steering torque sensor **91**, a steering angle sensor **92**, a vehicle body acceleration sensor **93**, a vehicle speed sensor **94**, a camera device **96**, and a radar device **97**.

(49) The steering torque sensor **91** is, for example, a magnetostrictive torque sensor provided between the handle **2** and the steering system component **10A** other than the handle **2**. The steering torque sensor **91** detects a torsional torque (steering input) input from the handle **2** to the other steering system components **10A**.

(50) The steering angle sensor **92** is, for example, a potentiometer provided on the steering shaft (stem shaft **34**). The steering angle sensor **92** detects the turning angle (steering angle) of the steering shaft with respect to the vehicle body.

(51) The vehicle body acceleration sensor **93** is a 5-axis or 6-axis inertial measurement unit (IMU). The vehicle body acceleration sensor **93** detects the angular velocities of the three axes (roll axis, pitch axis, and yaw axis) of the vehicle body and further can estimate the angle and acceleration from the results.

(52) The vehicle speed sensor **94** detects, for example, the rotation speed of the output shaft of the power unit **20**. The vehicle speed sensor **94** can detect the rotation speed of the rear wheel **4** and further the vehicle speed of the motorcycle **1** from the rotation speed.

(53) The camera device **96** includes a camera using a solid-state imaging device such as CCD or CMOS. The camera device **96**, for example, periodically photographs the surroundings of the motorcycle **1** (for example, front, rear, left, and right) using the camera. The camera device **96** generates image data from the captured image through image processing such as filtering and binarization.

(54) The radar device **97** radiates radio waves such as millimeter waves around the motorcycle **1**. The radar device **97** detects radio waves (reflected waves) reflected by an object around the vehicle. The radar device **97** can detect at least the front, rear, left, and right positions of the object with respect to the motorcycle **1** (distance and orientation with respect to the motorcycle **1**) and the

speed.

(55) Information from the camera device **96** and the radar device **97** described above is used to recognize the position, type, speed, and the like of an object in the detection direction. Based on this recognition, driving assistance control, automatic driving control, and the like of the motorcycle **1** are performed.

(56) Various devices **82** include the steering actuator **74**, the brake actuator **102**, the indicator lamp group **63**, and the movable portion actuator **105**.

(57) The steering actuator **74** generates a steering torque for steering the front wheel **3** independently of the operation of the handle **2** by the driver. The steering actuator **74** may also serve as a steering damper.

(58) The brake actuator **102** operates the front wheel brake **3B** and the rear wheel brake **4B** by supplying a hydraulic pressure to the front wheel brake **3B** and the rear wheel brake **4B** separately from the operation of the brake operator by the driver. The brake actuator **102** may also serve as an anti-lock brake system (ABS) control unit. The brake actuator **102** may be connected to a brake line branched from the normal brake circuit.

(59) The indicator lamp group **63** includes a right indicator lamp **66** and a left indicator lamp **67**. The right indicator lamp **66** and the left indicator lamp **67** emit light while being interlocked with the operation of the movable portion actuator **105**.

(60) The movable portion actuator **105** includes the right movable portion actuator **105R** and the left movable portion actuator **105L**. The right movable portion actuator **105L** drives the right movable portion **56** provided at the right portion of the vehicle body. The left movable portion actuator **105L** drives the left movable portion **57** provided at the left portion of the vehicle body.

(61) Next, the control device **71** will be described.

(62) The engine control unit **85** controls the output of the engine **13** based on the throttle opening, intake negative pressure, fuel injection amount, valve timing, ignition timing, and the like in the engine **13**. Further, the vehicle speed of the motorcycle **1** is changed according to the crankshaft rotation speed of the engine **13** and the gear ratio of the transmission **21** by controlling the output of the engine **13**.

(63) The steering control unit **77** controls the operation of the steering actuator **74** based on the following signals and information. The signals and information are the steering torque signal detected by the steering torque sensor **91**, the angular velocity signal detected by the vehicle body acceleration sensor **93**, the vehicle speed signal detected by the vehicle speed sensor **94**, the detection information detected by the camera device **96** and the radar device **97**, and the like. Accordingly, the assist torque is applied to the steering system component **10A**. By this assist torque, the steering of the front wheel **3** which is a steered wheel is assisted. In this way, the steering control unit **77** controls the automatic steering intervention function.

(64) The brake control unit **87** controls the operation of the brake actuator **102** based on the engine output, the vehicle speed signal detected by the vehicle speed sensor **94**, the detection information detected by the camera device **96** and the radar device **97**, and the like. Accordingly, the front wheel brake **3B** and the rear wheel brake **4B** generate an assist braking force. By this assist braking force, the braking of the front wheel **3** and the rear wheel **4** is assisted. In this way, the brake control unit **87** controls the automatic braking intervention function.

(65) The display control unit **88** controls the light emission (lighting or blinking) of the right indicator lamp **66** and the left indicator lamp **67** in accordance with the control of the following functions. The above-described function control includes the control of the automatic steering intervention function by the steering control unit **77** and the control of the automatic braking intervention function by the brake control unit **87**.

(66) The leg pressing control unit **89** controls the operation of the right movable portion actuator **105R** and the left movable portion actuator **105L** in accordance with the following function control. The function control includes the control of the automatic steering intervention function by the

steering control unit **77** and the control of the automatic braking intervention function by the brake control unit **87**. When the operation of the right movable portion actuator **105R** and the left movable portion actuator **105L** is controlled, the movement of the right movable portion **56** and the left movable portion **57** toward the outside in the vehicle width direction is controlled.

(67) The display control unit **88** and the leg pressing control unit **89** are synchronously controlled when the steering actuator **74** and the brake actuator **102** are operated.

(68) The engine control unit **85**, the steering control unit **77**, the brake control unit **87**, the display control unit **88**, and the leg pressing control unit **89** described above all include microcomputers and are configured to communicate with each other.

(69) <Leg Pressing Portion>

(70) Hereinafter, the configuration of the left leg pressing portion **72L** shown in FIGS. **5** to **7** will be described. Since the right leg pressing portion **72R** has a symmetrical configuration, the description thereof will be omitted.

(71) The left leg pressing portion **72L** includes the left movable portion actuator (actuator) **105L**, a disc **106**, a link member **107**, and the left movable portion **57**.

(72) The left movable portion actuator **105L** is an electric motor which is disposed between the upper edge of the main frame **7** and the fuel tank **18**. The left movable portion actuator **105L** is fixed to the main frame **7** or the fuel tank **18**. For example, the electric motor is a stepping motor which can control a rotation angle of a rotating shaft **105a** at a predetermined angle. For example, when the rotating shaft **105a** is stopped at a predetermined rotation angle, the electric motor has a holding force of holding the rotating shaft **105a** at that rotation angle. The disc **106** is attached to the rotating shaft **105a** of the left movable portion actuator **105** to be rotatable together. The rotating shaft **105a** extends in the front and rear direction of the vehicle.

(73) The upper end portion of the left movable portion **57** is supported by the lower end portion of a tank side cover **111**. The tank side cover **111** is disposed above the main frame **7**. The left movable portion **57** is swingably supported with an upper edge extending in the front and rear direction of the vehicle (extending along the upper edge of the main frame **7**) as a fulcrum. The outer side surface **57e** of the left movable portion **57** can contact the leg of the driver riding the motorcycle **1** from the inside in the vehicle width direction.

(74) The left movable portion **57** shown in FIG. **5** is at the position in which the left movable portion is swung most inward in the vehicle width direction. The position of the left movable portion **57** at this time is referred to as an initial position **P0** or a closed position.

(75) A link connection portion **57c** is provided on the inside of the left movable portion **57** in the vehicle width direction. The disc **106** is connected to the link connection portion **57c** through the link member **107**. One end portion of the link member **107** is swingably connected to the outer peripheral portion of the disc **106** by a first support shaft **113**. The other end portion of the link member **107** is swingably connected to the link connection portion **57c** of the left movable portion **57** by a second support shaft **114**.

(76) Additionally, the right movable portion actuator (actuator) **105R** of the right leg pressing portion **72R** is shown in the block diagram of FIG. **4**. The right movable portion actuator **105R** is an electric motor having the same structure as that of the left movable portion actuator **105L**. The right movable portion **57** has the outer side surface **56e** that is able to contact the leg of the driver riding on the vehicle.

(77) As shown in FIG. **6**, when the left movable portion actuator **105L** is operated from the state of FIG. **5** in the left leg pressing portion **72L**, the following operations occur. That is, the disc **106** rotates in the direction of the arrow **A** together with the rotating shaft **105a** of the left movable portion actuator **105L**.

(78) In accordance with this rotation, the link member **107** is pressed outward in the vehicle width direction so that the left movable portion **57** is pressed outward in the vehicle width direction. As a result, the left movable portion **57** swings with the upper end portion as the swing fulcrum and

moves as below. That is, the left movable portion **57** moves from the initial position **P0** (closed position) indicated by the two-dotted chain line in FIG. **5** to the first operation position **P1** (LOW position) indicated by the solid line. The left movable portion **57** protrudes outward in the vehicle width direction by a first prescribed amount **S1** when swinging from the initial position **P0** to the first operation position **P1**.

(79) As shown in FIG. **7**, when the left movable portion actuator **105L** is further operated from the state of FIG. **6** in the left leg pressing portion **72L**, the following operation occurs. That is, the disc **106** further rotates in the direction of the arrow **A** together with the rotating shaft **105a** of the left movable portion actuator **105L**.

(80) In accordance with this rotation, the link member **107** is further pressed outward in the vehicle width direction so that the left movable portion **57** is further pressed outward in the vehicle width direction. As a result, the left movable portion **57** further swings with the upper end portion as the swing fulcrum and moves as follow. That is, the left movable portion **57** moves from the first operation position **P1** (LOW position) indicated by the two-dotted chain line in FIG. **6** to the second operation position **P2** (HIGH position) indicated by the solid line. The left movable portion **57** protrudes outward in the vehicle width direction by a second prescribed amount **S2** when swinging from the first operation position **P1** to the second operation position **P2**. For example, the second operation position **P2** is a position in which the left movable portion **57** protrudes maximum outward in the vehicle width direction.

(81) The left movable portion **57** (and the right movable portion **56**) can move in two stages of the LOW position in which the outward protrusion amount in the vehicle width direction is small and the HIGH position in which the outward protrusion amount in the vehicle width direction is large. Accordingly, it is possible to notify the driver of a plurality of different types of information regarding the automatic operation intervention function.

(82) Additionally, the operation of the left movable portion **57** (and the right movable portion **56**) to give a tactile stimulus to the driver's legs is not limited to swinging with the upper end portion as the swinging fulcrum. For example, the left movable portion **57** (and the right movable portion **56**) may swing with the lower end portion and the front and rear end portions as swing fulcrums. Further, the left movable portion **57** (and the right movable portion **56**) may generally translate as a whole, perform various other operations, vibrate the outer side surfaces **56e** and **57e**, or combine a plurality of the various operations described above.

(83) The operation of the left and right leg pressing portions **72L** and **72R** described above will be described.

(84) First, when the system of the driving assistance device **70** is not activated, the right movable portion **56** and the left movable portion **57** are arranged at the initial position **P0** as shown in FIG. **5**. That is, the right movable portion **56** and the left movable portion **57** are arranged along the outer surfaces of the right vehicle body outer portion **1R** and the left vehicle body outer portion **1L** of the motorcycle **1**. The right movable portion **56** and the left movable portion **57** at the initial position **P0** are in the following states in the right vehicle body outer portion **1R** and the left vehicle body outer portion **1L**. That is, the right movable portion **56** and the left movable portion **57** form a substantially continuous outer surface together with vehicle body components such as the main frame **7**, the fuel tank **18**, and other vehicle body covers.

(85) The right movable portion **56** and the left movable portion **57** are located at a position (so-called knee grip position) in which the driver sandwiches the vehicle body **1A** at the left and right legs. When the driving assistance device **70** is not activated, the right movable portion **56** and the left movable portion **57** do not protrude on the outer surfaces of the right vehicle body outer portion **1R** and the left vehicle body outer portion **1L**. Therefore, the driver can drive comfortably without being particularly aware of the presence of the right movable portion **56** and the left movable portion **57** when the right vehicle body outer portion **1R** and the left vehicle body outer portion **1L** perform a knee gripping operation.

(86) Next, a non-operation state in which the system of the driving assistance device **70** is activated and the automatic operation intervention function is not operated yet (hereinafter, referred to as a standby state) will be described. At this time, as shown in FIG. **6**, the right movable portion **56** and the left movable portion **57** are arranged to protrude outward in the vehicle width direction by the first prescribed amount **S1** from the outer surfaces of the right vehicle body outer portion **1R** and the left vehicle body outer portion **1L** of the motorcycle **1**.

(87) In the standby state, the right movable portion **56** and the left movable portion **57** repeat the reciprocating movement between the initial position **P0** and the first operation position **P1**. The first operation position **P1** is a position that protrudes outward in the vehicle width direction by the first prescribed amount **S1** from the initial position **P0**. Therefore, the driver is aware of the presence of the right movable portion **56** and the left movable portion **57** on the left and right legs that perform a knee gripping operation on the right vehicle body outer portion **1R** and the left vehicle body outer portion **1L**. Therefore, the driver can recognize that the driving assistance device **70** is activated.

(88) Next, the operation states of the steering intervention function and the braking intervention function of the automatic operation intervention function from the standby state of the driving assistance device **70** will be described. At this time, as shown in FIG. **7**, at least one of the right movable portion **56** and the left movable portion **57** is disposed to protrude further outward in the vehicle width direction from the first operation position **P1**. At this time, the right movable portion **56** and the left movable portion **57** are located at the second operation position **P2** that protrudes further outward in the vehicle width direction by the second prescribed amount **S2** from the first operation position **P1**.

(89) Accordingly, the driver can recognize that the automatic operation intervention function is operated by detecting the movement of at least one of the right movable portion **56** and the left movable portion **57**.

(90) In the embodiment, at least one of the right movable portion **56** and the left movable portion **57** is maintained in a state having moved to the second operation position **P2** when the automatic operation intervention function is operated. However, the driving assistance device is not limited to this configuration. For example, at least one of the right movable portion **56** and the left movable portion **57** may repeat the reciprocating movement between the first operation position **P1** and the second operation position **P2** when the automatic operation intervention function is operated.

(91) The operation of the driving assistance device **70** described above will be described.

(92) FIG. **8** shows a standby state in which the system of the driving assistance device **70** is activated and the automatic operation intervention function is not operated yet. In the standby state, the main switch (power supply) of the motorcycle **1** is turned on and the operation switch of the driving assistance device **70** is turned on so that the operation of the automatic operation intervention function is allowed. In the standby state, various sensors **81** monitor various parameters and the like for operating the automatic operation intervention function. Based on this monitoring information, the automatic operation intervention function can be operated immediately.

(93) In the standby state, the right movable portion **56** of the right leg pressing portion **72R** and the left movable portion **57** of the left leg pressing portion **72L** are operated in a synchronized manner. At this time, the right movable portion **56** and the left movable portion **57** perform one reciprocating movement (swing) in the vehicle width direction. Specifically, the right movable portion **56** and the left movable portion **57** perform the forward movement from the initial position **P0** (closed position) to the first operation position **P1** (LOW position) and the backward movement from the first operation position **P1** to the initial position **P0**.

(94) Further, at this time, the right indicator lamp **66** and the left indicator lamp **67** of the meter device **61** emit light (blinking) in synchronization with the operation of the right movable portion **56** and the left movable portion **57**. Specifically, the right indicator lamp **66** and the left indicator lamp **67** emit light when the right movable portion **56** and the left movable portion **57** reach the

first operation position **P1** from the initial position **P0**. Further, the right indicator lamp **66** and the left indicator lamp **67** are extinguished when returning from the first operation position **P1** to the initial position **P0**. The blinking period of the right indicator lamp **66** and the left indicator lamp **67** in the standby state (which is also the operation period of the left and right movable portions **56** and **57**) is set as a first period.

(95) In this way, a tactile stimulus is applied to the driver's legs and a visual sense is also used to notify the driver that the system is activated.

(96) FIG. **9** shows the time course of the operation of the movable portion actuator **105** and the like in the standby state.

(97) The vertical axis of the graph of FIG. **9** indicates the operation states (ON (operation) and OFF (non-operation)) of the actuators **105L** and **105R** of the left and right movable portions **56** and **57** and the left and right indicator lamps **66** and **67**. The vertical axis also indicates the operating state (ON and OFF) of the steering or braking intervention.

(98) In the operation state of the left and right actuators **105L** and **105R**, a case in which the left and right movable portions **56** and **57** are located at the closed position is an OFF state and a case in which the left and right movable portions are located at the LOW position and the HIGH position is an ON state. Further, a case in which the left and right movable portions **56** and **57** are located at the LOW position is indicated by "L" in the graph and a case in which the left and right movable portions are located at the HIGH position is indicated by "H" in the graph. The horizontal axis of the graph indicates the time **T** (sec).

(99) In the drawing, "RH movable portion" indicates the right movable portion actuator **105R** and "LH movable portion" indicates the left movable portion actuator **105L**, respectively. Further, "indRH" indicates the right indicator lamp **66** and "indLH" indicates the left indicator lamp **67**, respectively. Additionally, also in the graphs of FIGS. **11**, **13**, and **15**, the vertical axis and the horizontal axis are the same as those of the graph shown in FIG. **9**.

(100) In the above-described standby state, the right movable portion actuator **105R** and the left movable portion actuator **105L** are intermittently operated at the same timing. In the standby state, the right movable portion actuator **105R** and the left movable portion actuator **105L** perform an operation to open the left and right movable portions **56** and **57** from the initial position **P0** to the first operation position **P1**. This operation is performed only for a predetermined time **T1** (for example, 0.5 to 1 second). In the standby state, the right movable portion actuator **105R** and the left movable portion actuator **105L** perform an operation of opening the left and right movable portions **56** and **57**. This operation is repeated at intervals of a predetermined time **T2** (for example, about 2 seconds).

(101) The right movable portion **56** and the left movable portion **57** which are opened outward in the vehicle width direction from the initial position **P0** contact and press the left and right legs (thighs, knees, and the like) of the driver from the inside in the vehicle width direction. In the standby state, the right movable portion **56** and the left movable portion **57** move to the LOW position protruding from the surrounding leg grip portion **120**. At this time, the right movable portion **56** and the left movable portion **57** open the left and right legs of the driver outward in the vehicle width direction rather than being positioned at the closed position. As a result, the driver recognizes the standby state from the operation of the right movable portion **56** and the left movable portion **57**.

(102) At this time, the right movable portion **56** and the left movable portion **57** move outward in the vehicle width direction and press the left and right legs of the driver, so that the reaction increases the force with which the driver sandwiches the vehicle body. Further, at this time, an action occurs in which the driver strongly sandwiches the vehicle body against the pressure. As a result, the left and right legs of the driver are able to hold the vehicle body more firmly and the riding posture is suitable for the automatic operation intervention function.

(103) Further, the right indicator lamp **66** and the left indicator lamp **67** blink in synchronization

with the intermittent operation of the right movable portion **56** and the left movable portion **57**. That is, the right indicator lamp **66** and the left indicator lamp **67** repeat the light emission of the time **T1** with a time interval of the time **T2**. In this way, the right movable portion **56** and the left movable portion **57** are interlocked with the right indicator lamp **66** and the left indicator lamp **67**.

(104) In the standby state, the steering intervention function and the braking intervention function of the automatic operation intervention function are not operated and are in an OFF state.

(105) FIG. **10** shows a state in which the driving assistance device **70** operates the automatic braking intervention function from the standby state to brake the motorcycle **1**.

(106) Since the automatic braking intervention function of the embodiment has no relation in the left and right direction, both the right movable portion **56** and the left movable portion **57** are operated. At this time, both the right movable portion **56** and the left movable portion **57** are moved from the closed position to the HIGH position. Since it is assumed that the automatic braking intervention function provokes the physical behavior of the driver in an urgent state, both the right movable portion **56** and the left movable portion **57** are operated. At this time, the right movable portion **56** and the left movable portion **57** are not intermittently operated, but continuously operated without time intervals.

(107) Further, both the right indicator lamp **66** and the left indicator lamp **67** emit light (blink) while the right movable portion **56** and the left movable portion **57** are operated. This light emission is not limited to blinking and may be lighting interlocked with the operation of the right movable portion **56** and the left movable portion **57**.

(108) When the above-described automatic braking intervention function is operated, the right movable portion **56** and the left movable portion **57** are operated relatively largely in an operation mode different from that in the standby state. Further, the right indicator lamp **66** and the left indicator lamp **67** of the meter device **61** blink faster than in the standby state.

(109) Accordingly, the driver can recognize that the automatic braking intervention function is operated through the visual sense and the tactile sense of the left and right legs that perform a knee gripping operation on the vehicle body.

(110) In the operation state of the automatic braking intervention function, the right movable portion **56** and the left movable portion **57** move to the HIGH position protruding from the surrounding leg grip portion **120**. Accordingly, the right movable portion **56** and the left movable portion **57**, which protrude more than the LOW position, open the left and right legs wider outward in the vehicle width direction than at the LOW position. Accordingly, the driver strongly recognizes that the automatic braking intervention function is operated.

(111) At this time, since both legs are pressed by the right movable portion **56** and the left movable portion **57** from the inside, the following effects occur. That is, the holding force of holding the vehicle body by both legs is higher than that at the LOW position. Accordingly, the riding posture is suitable for coping with the deceleration of the motorcycle **1** when the automatic braking intervention function is operated.

(112) FIG. **11** shows the operation mode of the right movable portion actuator **105R** and the left movable portion actuator **105L** when the automatic braking intervention function is operated.

(113) When the automatic braking intervention function is operated by an obstacle or the like in front of the motorcycle **1**, both the right movable portion actuator **105R** and the left movable portion actuator **105L** start to be operated. At this time, the right movable portion **56** and the left movable portion **57** move from the closed position to the HIGH position. This operation is maintained continuously without a time interval during the time **T5** during which the automatic braking intervention function is operated.

(114) Further, both the right indicator lamp **66** and the left indicator lamp **67** start blinking during the operation of the right movable portion **56** and the left movable portion **57**. Specifically, the right indicator lamp **66** and the left indicator lamp **67** repeat light emission for a predetermined time **T3** (for example, 0.5 to 1 second). The light emission at the time **T3** is repeated at intervals of a

predetermined time **T4** (for example, 0.5 to 1 second). The time **T4** is shorter than the time **T2** in the standby state. The time **T3** may be equal to or shorter than the time **T1** in the standby state.

(115) Here, a timing **te** in which the right movable portion **56** and the left movable portion **57** start to be operated and the right indicator lamp **66** and the left indicator lamp **67** start to blink is set as below. That is, the timing **te** is earlier than a timing **tf** when the automatic braking intervention function starts to be operated by a time **th** (for example, about 1 second). That is, the operating of the right movable portion **56** and the left movable portion **57** and the blinking of the right indicator lamp **66** and the left indicator lamp **67** start before the operation of the automatic braking intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the automatic braking intervention function.

(116) Additionally, the operating of the right movable portion **56** and the left movable portion **57** and the blinking of the right indicator lamp **66** and the left indicator lamp **67** are simultaneously stopped at a timing **tg** when the operation of the automatic steering intervention function is stopped.

(117) FIG. **12** shows a state in which the driving assistance device **70** operates the automatic steering intervention function from the standby state and the course of the motorcycle **1** is about to turn to the left side.

(118) For example, when the steering control unit **77** of the driving assistance device **70** determines that the motorcycle **1** has moved too far to the right side of the lane, the steering control unit **77** tries to return the course of the motorcycle **1** to the left side.

(119) At this time, the steering control unit **77** steers the front wheel **3** to the right side by the steering actuator **74**. That is, the front wheel **3** is steered in the opposite direction to the side on which the course of the motorcycle **1** is to be changed (reverse steering). Accordingly, the motorcycle **1** can be naturally steered in the course change direction.

(120) When the above-described automatic steering intervention function is operated, only the left movable portion **57** in the course change direction (left side) of the motorcycle **1** moves from the LOW position to the HIGH position. Further, the left indicator lamp **67** of the meter device **61** blinks in a blinking mode different from that in the standby state.

(121) Accordingly, the driver can recognize the following points through the visual sense and the tactile sense of the legs that perform the knee gripping operation on the vehicle body. That is, the driver can recognize the movement direction of the motorcycle **1** since the automatic steering intervention function is operated.

(122) When the left movable portion **57** presses the occupant's left leg outward (leftward) in the vehicle width direction, the left leg is opened outward (leftward) in the vehicle width direction. Accordingly, the driver can strongly recognize that the automatic steering intervention function is operated to the left side of the vehicle. Further, when the left leg is opened to the left side, the driver's center of gravity moves to the left side with respect to the vehicle body and the driver can take a posture suitable for left turning.

(123) FIG. **13** shows the operation mode of the right movable portion actuator **105R** and the left movable portion actuator **105L** during the operation shown in FIG. **12** of the automatic steering intervention function.

(124) During the operation shown in FIG. **12** of the automatic steering intervention function, only the left movable portion actuator **105L** is operated so that the left movable portion **57** is moved from the LOW position to the HIGH position.

(125) During the above-described operation, the left indicator lamp **67** starts to blink differently from the standby state. This blinking may be, for example, in a mode of repeating the light emission of the time **T3** (for example, 0.5 to 1 second) with a time interval of the time **T4** (for example, 0.5 to 1 second) as in the case of the automatic braking intervention.

(126) Here, a timing **ta** when the left movable portion **57** moves to the HIGH position and the left indicator lamp **67** starts to blink is set as below. That is, the timing **ta** is earlier than a timing **tb**

when the automatic steering intervention function starts to be operated by a time t_d (for example, about 1 second). That is, the operating of the left movable portion actuator **105L** and the blinking of the left indicator lamp **67** are started before the operation of the automatic steering intervention function. Accordingly, the driver can be notified of the start of the operation of the automatic steering intervention function.

(127) Additionally, the operating of the left movable portion actuator **105L** and the blinking of the left indicator lamp **67** are simultaneously stopped at a timing t_c when the operation of the automatic steering intervention function is stopped.

(128) FIG. **14** shows a state in which the driving assistance device **70** activates the automatic steering intervention function from the standby state and is about to turn the course of the motorcycle **1** to the right side.

(129) For example, when the steering control unit **77** of the driving assistance device **70** determines that the motorcycle **1** has moved too far to the left side of the lane, the steering control unit **77** tries to return the course of the motorcycle **1** to the right side.

(130) At this time, the steering control unit **77** steers the front wheel **3** to the left side by the steering actuator **74**. That is, the front wheel **3** is steered in the opposite direction to the side on which the course of the motorcycle **1** is to be changed (reverse steering). Accordingly, the motorcycle **1** can be naturally steered in the course change direction.

(131) When the automatic steering intervention function is operated, only the right movable portion **56** in the course change direction (right side) of the motorcycle **1** moves from the LOW position to the HIGH position. Further, the right indicator lamp **66** of the meter device **61** blinks in a blinking mode different from that in the standby state.

(132) Accordingly, the driver can recognize the following points through the visual sense and the tactile sense of the legs that perform the knee gripping operation on the vehicle body. That is, it is possible to recognize the movement direction of the motorcycle **1** by the operation of the automatic steering intervention function.

(133) When the right movable portion **56** presses the occupant's right leg outward (rightward) in the vehicle width direction, the right leg is opened outward (rightward) in the vehicle width direction. Accordingly, the driver can strongly recognize that the automatic steering intervention function is operated to the right side of the vehicle. Further, when the right leg is opened to the right side, the driver's center moves to the right side with respect to the vehicle body and the driver can take a posture suitable for right turning.

(134) FIG. **15** shows the operation mode of the right movable portion actuator **105R** and the left movable portion actuator **105L** during the operation shown in FIG. **14** of the automatic steering intervention function.

(135) During the operation shown in FIG. **14** of the automatic steering intervention function, only the right movable portion actuator **105R** is operated so that the right movable portion **56** is moved from the LOW position to the HIGH position.

(136) During the above-described operation, the right indicator lamp **66** starts to blink differently from the standby state. This blinking may be, for example, in a mode of repeating the light emission of the time T_3 (for example, 0.5 to 1 second) with a time interval of the time T_4 (for example, 0.5 to 1 second) as in the case of the automatic braking intervention.

(137) Here, a timing t_a when the right movable portion **56** moves to the HIGH position and the right indicator lamp **66** starts to blink is set as below. That is, the timing t_a is earlier than a timing t_b when the automatic steering intervention function starts to be operated by a time t_d (for example, about 1 second). That is, the operating of the right movable portion actuator **105R** and the blinking of the right indicator lamp **66** are started before the operation of the automatic steering intervention function. Accordingly, the driver can be notified of the start of the operation of the automatic steering intervention function.

(138) Additionally, the operating of the right movable portion actuator **105R** and the blinking of the

right indicator lamp **66** are simultaneously stopped at a timing t_c when the operation of the automatic steering intervention function is stopped.

(139) As described above, in the vehicle information notification device of the above-described embodiment, at least one of the pair of left and right vehicle body outer portions **1L** and **1R** in the vehicle body **1A** steerably supporting the steering system component **10A** is provided with the left and right movable portions **56** and **57** which are operated by the left and right movable portion actuators **105L** and **105R**. The left and right movable portions **56** and **57** include the left and right outer side surfaces **56e** and **57e** which are provided in front of the seat **19** on which the driver sits and above the step **25** on which the driver's feet are placed so that the driver's legs are able to contact the left and right outer side surfaces. The left and right movable portions **56** and **57** notify the driver of predetermined information by operating the left and right outer side surfaces **56e** and **57e** in a predetermined motion.

(140) According to this configuration, at least one of the pair of left and right vehicle body outer portions **1L** and **1R** of the vehicle body **1A** is provided with the left and right movable portions **56** and **57**. By the operation of the left and right movable portions **56** and **57**, predetermined information can be notified to the driver's legs. The left and right movable portions **56** and **57** are separated from the steering system component **10A** (especially, the handle **2**). Accordingly, the notification of information by the left and right movable portions **56** and **57** and the steering operation are separated from each other and the driver can easily receive notified information.

(141) Further, in the vehicle information notification device, the left and right movable portions **56** and **57** are operated so that the left and right outer side surfaces **56e** and **57e** are moved outward in the vehicle width direction. Accordingly, it is possible to increase the certainty of contact with the driver's legs and to easily receive notified information by the driver.

(142) Further, in the vehicle information notification device, the left and right movable portions **56** and **57** are respectively provided in the pair of left and right vehicle body outer portions **1L** and **1R** and are operable independently of each other. Accordingly, it is possible to combine the difference in which of the left and right movable portions **56** and **57** is operated. Accordingly, it is possible to notify the driver of a lot of information and to increase the type of information to be notified.

(143) Further, the vehicle information notification device includes the operation intervention function capable of intervening in operations performed by the driver and the left and right movable portions **56** and **57** are operated while being interlocked with the operation intervention function. Accordingly, it is possible to notify the driver that the operation intervention function is operated by the operation of the left and right movable portions **56** and **57**.

(144) Further, in the vehicle information notification device, when the operation intervention function is operated, the left and right movable portions **56** and **57** start to be operated at the timings t_a and t_e earlier than the timings t_b and t_f of starting the operation of the operation intervention function. Accordingly, it is possible to notify the driver of the start of the operation of the operation intervention function in advance.

(145) Further, in the vehicle information notification device, at least one of the steering intervention function and the braking intervention function is provided as the operation intervention function. Accordingly, it is possible to notify the driver that at least one of the steering intervention function and the braking intervention function is operated by the operation of the left and right movable portions **56** and **57**.

(146) Further, in the vehicle information notification device, both left and right movable portions **56** and **57** are operated when the braking intervention function is operated. Accordingly, it is possible to notify the driver that the braking intervention function is operated by the operation of both left and right movable portions. Accordingly, it is possible to increase the certainty of the notification of the operation of the braking intervention function that tends to cause behavior.

(147) Further, in the vehicle information notification device, one of the left and right movable portions **56** and **57** is operated when the steering intervention function is operated. Accordingly, it

is possible to notify the driver that the steering intervention function is operated by the operation of one of the left and right movable portions **56** and **57**. That is, it is possible to notify the driver of information including the steering direction by the steering intervention function depending on which of the left and right movable portions **56** and **57** is operated.

(148) Further, in the vehicle information notification device, the steering intervention function includes a course change function for changing the course of the motorcycle **1** to any one of left and right. When the steering intervention function is operated, one that changes the course of the motorcycle **1** in the left and right movable portions **56** and **57** is operated so that the outer side surfaces **56e** and **57e** are moved outward in the vehicle width direction. Accordingly, the leg of the driver on the side of the course change is opened outward in the vehicle width direction, and the posture suitable for steering can be urged.

(149) Further, in the vehicle information notification device, the left and right movable portions **56** and **57** are operated as below in the standby state in which the system of the operation intervention function is activated and the operation intervention function is not operated yet. That is, the left and right movable portions **56** and **57** are operated in an operation mode different from that when the operation intervention function is operated. Accordingly, it is possible to notify the driver that the operation intervention function is in the standby state.

(150) Further, the vehicle information notification device further includes the left and right display devices **66** and **67** which display predetermined information for the driver while being interlocked with the operation of the left and right movable portions **56** and **57**. Accordingly, it is possible to further reliably notify the driver of information by the operation of the left and right movable portions **56** and **57** and the display of the left and right display devices **66** and **67**.

(151) Additionally, the present invention is not limited to the above-described embodiment. For example, the driving assistance device **70** includes the steering intervention function and the braking intervention function as the operation intervention function, but the present invention is not limited thereto. That is, the driving assistance device **70** may include only one of the steering intervention function and the braking intervention function as the operation intervention function.

(152) In the embodiment, an example of notifying the operation information of the device related to driving assistance was shown, but the present invention is not limited thereto. That is, the notification may be, for example, the notification of information from a navigation system or ITS (Intelligent Transport Systems).

(153) The present invention is not limited to a configuration in which the movable portion is provided in both left and right vehicle body outer portions. That is, the present invention may have a configuration in which the movable portion is provided in only one of the left and right vehicle body outer portions.

(154) The movable portion may be provided in a part (including the step) of the vehicle body outer portion where the driver's lower legs and feet (for example, calves, ankles, soles, and the like) can come into contact. The step is not limited to a bar type, but include a floor type and a board type. The movable portion may be provided in the seat that contacts the driver's buttocks or around the base of the leg or in the lumbar that contacts the waist. That is, the movable portion may form an outer surface that naturally comes into contact with the body (especially, the lower half of the body) of the driver in the normal riding posture.

(155) The saddle type vehicle includes all vehicles in which the driver straddles the vehicle body. That is, the saddle type vehicle includes not only motorcycles (including motorized bicycles and scooter-type vehicles), but also vehicles with three wheels (including vehicles with one front wheel and two rear wheels as well as vehicles with two front wheels and one rear wheel) or four wheels. Further, the saddle type vehicle also includes vehicles whose prime mover includes an electric motor.

(156) Then, the configuration of the above-described embodiment is an example of the present invention. That is, various modifications can be made without departing from the gist of the present

invention, such as replacing the constituent elements of the embodiments with well-known constituent elements.

BRIEF DESCRIPTION OF THE REFERENCE SYMBOLS

(157) **1**: Motorcycle (saddle type vehicle) **1A**: Vehicle body **1L**: Left vehicle body outer portion (vehicle body outer portion) **1R**: Right vehicle body outer portion (vehicle body outer portion) **10A**: Steering system component **19**: Seat **25**: Step **56**: Right movable portion (movable portion) **56e**: Outer side surface **57**: Left movable portion (movable portion) **57e**: Outer side surface **66**: Right indicator lamp (display device) **67**: Left indicator lamp (display device) **105R**: Right movable portion actuator (actuator) **105L**: Left movable portion actuator (actuator) tb, tf: Timing (timing of starting operation of operation intervention function) ta, te: Timing (timing earlier than start of operation of operation intervention function)

Claims

1. A motorcycle information notification device comprising: movable portions which are provided in at least one of a pair of left and right vehicle body outer portions in a vehicle body and are operated by movable portion actuators, steerably supporting a steering system component; and an operation intervention function capable of intervening in operations performed by a driver, wherein the movable portion actuators are connected to the movable portions through a link member, wherein the movable portions include outer surfaces which are able to contact driver's legs, wherein the movable portions notify the driver of predetermined information by operating the outer surfaces in a predetermined motion, wherein the movable portions are operated while being interlocked with the operation intervention function, and wherein, when the operation intervention function is operated, the outer surfaces protrude outward in a vehicle width direction from remaining general surfaces of the vehicle body outer portions at a timing earlier than a timing of starting the operation of the operation intervention function.
2. The motorcycle information notification device according to claim 1, wherein the movable portions include the outer surfaces provided in front of a seat and above a step to be able to contact the driver's legs.
3. The motorcycle information notification device according to claim 1, wherein the movable portions are respectively provided in the pair of left and right vehicle body outer portions and are operable independently of each other.
4. The motorcycle information notification device according to claim 1, wherein at least one of a steering intervention function and a braking intervention function is provided as the operation intervention function.
5. The motorcycle information notification device according to claim 4, wherein both the left and right movable portions are operated when the braking intervention function is operated.
6. The motorcycle information notification device according to claim 4, wherein one of the left and right movable portions is operated when the steering intervention function is operated.
7. The motorcycle information notification device according to claim 4, wherein the steering intervention function includes a course change function for changing a course of a vehicle to any one of left and right, wherein the movable portions are operated so that the outer surfaces are moved outward in the vehicle width direction, and wherein when the steering intervention function is operated, the movable portions that change the course of the vehicle are operated so that the outer surfaces are moved outward in the vehicle width direction.
8. The motorcycle information notification device according to claim 1, wherein in a standby state in which a system of the operation intervention function is activated and the operation intervention function is not operated yet, the movable portions are operated in an operation mode different from that when the operation intervention function is operated.
9. The motorcycle information notification device according to claim 1, further comprising: display

devices which display predetermined information for the driver while being interlocked with the operation of the movable portions.

10. The motorcycle information notification device according to claim 1, wherein operation positions of the movable portions include at least a first operation position in which the outer surfaces are moved outward in the vehicle width direction by a first prescribed amount from an initial position, and a second operation position in which the outer surfaces are further moved outward in the vehicle width direction by a second prescribed amount from the first operation position.
